

CVATS – AI Powered ATS Resume Scanner and Job-Readiness Assistant

Project Proposal for UCS503P
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1 Higher-order goal

This project aims to improve job readiness and career mobility for job seekers by removing ambiguity from the resume screening and interview preparation process. While applicant tracking systems (ATS) are widely used across industries, the immediate engineering objective is to translate explainable AI resume matching and preparation into a measurable, deployable software solution.

2 Time-to-value

- We will deliver meaningful increments quickly by prototyping the core parsing and ATS match scoring workflow, validating it with sample candidate resumes within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations (and targeting automated testing early) to continuously integrate gap analysis, resume enhancement, and mock interview features.
- Success is defined by what can be delivered and measured in the first iteration (explainable match score), then expanded based on user feedback.

3 Problem Statement

Most job seekers today apply through Applicant Tracking Systems (ATS) without any real visibility into how their resume is actually being read by the software or how closely it matches the specific role they are applying for. Common pain points include:

- **Terminology mismatch:** Candidates may be well qualified on paper, yet get filtered out simply because their resume uses different terminology than the job description or omits required keywords.
- **Generic scoring:** Freely available “resume checker” tools stop at giving an opaque generic score or basic formatting tips without explaining the score or tying feedback to a specific job description.
- **Lack of actionable gap analysis:** Existing scanners answer “is my resume ATS-friendly?” but not “how well do I match this particular job, what is missing, and how do I close that gap?”
- **Disconnected post-shortlisting preparation:** Candidates prepare for interviews with generic question banks that have no connection to the role or their own background.

Why this matters now (contextual relevance): In a competitive job market with automated recruitment filters, providing explainable matching, non-fabricating resume enhancement, and role-specific mock interviews significantly reduces preparation friction and candidate drop-off.

4 Proposed Solution

4.1 Overview

Build a **web-based** “AI-Powered ATS Resume Scanner and Job Application Assistant” with the following components:

- **Resume & JD parsing portal:** Users upload PDF/DOCX resumes and paste target job descriptions.
- **Explainable ATS match engine:** Computes category-wise scores (skills, experience, keywords, education).
- **Gap analysis & recommendation module:** Separates “things to improve” from “things to learn/add”.
- **AI resume enhancement module:** Rewrites bullet points for clarity and keyword alignment without fabricating experience.
- **Role-specific mock interview system:** Conducts AI interview sessions based on the resume and JD, generating an actionable feedback report.

4.2 Core workflow

1. Candidate uploads resume (PDF/DOCX) and inputs a target job description.
2. System parses sections and performs hybrid matching (keyword matching + semantic embeddings) to generate an explainable match score with category breakdowns.
3. System presents gap analysis: weak content to improve vs. missing skills to learn.
4. Candidate generates an enhanced resume version with rewritten bullet points for review.
5. Candidate initiates a role-specific mock interview session, receives real-time questions, submits answers, and views a final interview performance report.

4.3 Operational constraints for an educational, tier-2/3 setting

- Keep the solution usable on low-to-mid range devices via responsive web design (React/Next.js).
- Avoid dependence on complex custom ML models; focus on practical LLM pipeline design, prompt engineering, and system correctness.
- Design for deployability using common cloud web hosting and managed relational databases (PostgreSQL/MySQL).

5 Solution Approach

This is an applied engineering project centered on a robust LLM application pipeline rather than novel algorithm research.

5.1 Duplicate/quality assistance

Resume-JD matching and gap identification will be engineering-guided:

- Normalize extracted text across standard sections (Skills, Experience, Education, Projects).
- Use a hybrid matching approach (keyword extraction combined with LLM semantic embeddings) to break down sub-scores.
- Apply strict prompt constraints and validation checks during resume enhancement to guarantee zero fabrication of employers, tools, or metrics.

5.2 Web architecture

A standard 3-tier web architecture:

- **Frontend:** React / Next.js responsive UI for resume upload, analysis dashboard, bullet rewriter, and mock interview interface.
- **Backend API:** Node.js / Express.js (or Python FastAPI) orchestrating text extraction, LLM API calls, scoring logic, and session persistence.
- **Database:** Relational database (PostgreSQL/MySQL) storing user profiles, uploaded resumes, JDs, analysis records, and interview logs.

5.3 Automated testing and fast delivery enabler

Automated testing will be used to:

- Automatically execute unit tests for parsing, score calculations, and prompt template validation.
- Produce repeatable build artifacts for deployment to staging environments.
- Enable safe and frequent releases of incremental pipeline modules.

6 Evaluation Criterion

We define success using measurable metrics tied to matching explainability and user preparation efficiency.

6.1 Primary evaluation metric

Match Score Explainability & Accuracy: Consistency of category-wise sub-scores (skills, keywords, experience) when evaluated against manually annotated sample resume-JD pairs.

- **Target:** match score correlation $\geq 85\%$ with human recruiter assessments on test benchmark.
- **Attribution:** measured by comparing system category sub-scores against expert evaluation rubrics.

6.2 Secondary evaluation metrics

- **Zero Fabrication Rate:** 100% adherence in the resume enhancement module (no invented companies, titles, or metrics).
- **Mock Interview Relevance:** rating of AI-generated questions against the candidate's actual resume and JD gaps.
- **Processing Latency:** end-to-end analysis generated in ≤ 10 seconds.
- **Reliability:** system availability $\geq 99\%$ during user testing iterations.

6.3 Pilot validation plan

- Recruit 10–20 student job seekers and test against diverse candidate resumes and job descriptions.
- Compare preparation outcomes before and after system usage through candidate feedback surveys and mock interview score progression.

7 Scalability

The system is designed to scale efficiently across both technical and operational dimensions.

1. **Scalable (theoretically):** The system supports growth in concurrent users and large resume document sets by:
 - decoupling frontend and backend services via stateless REST APIs,
 - caching analysis results for repeated resume-JD evaluations,
 - asynchronous background processing for heavy LLM operations.
2. **Operationally deployable (practically):** The platform deploys seamlessly using standard Web/Cloud infrastructure:
 - managed relational database instances,
 - platform-as-a-service (PaaS) hosting for frontend and API backend,
 - automated deployment scripts for rapid staging and deployment.

8 Engine availability heuristic

A mature set of software engines and AI services is available to implement this web project:

- Text extraction engines for PDF and DOCX documents (pdf-parse, mammoth).
- LLM APIs (OpenAI / Gemini / Anthropic) for parsing, gap analysis, bullet rewriting, and mock interviews.
- Modern web framework (React / Next.js, Node.js / Express.js / FastAPI).
- Managed relational database (PostgreSQL / MySQL).
- Automated test runner and build tools (GitHub Actions / Jest).

We leverage these mature tools to focus on system integration, prompt engineering, user experience, and evaluation rather than reinventing core infrastructure.

9 Project scope and deliverables

9.1 Initial deliverable

- Resume (PDF/DOCX) and Job Description text ingestion portal.
- Resume & JD parsing engine with section structuring.
- Explainable ATS match score engine with category-wise breakdown dashboard.
- Basic automated test suite for parsing and scoring accuracy.
- Single-command deployment to staging.

9.2 Subsequent deliverables

- Gap analysis module producing “Things to Improve” vs. “Things to Learn”.
- AI Resume Enhancement module with before/after comparison interface and strict validation.
- Role-specific AI Mock Interview module with answer evaluation and feedback report.
- Admin and candidate history dashboard for tracking progress across multiple applications.

10 Risks and mitigations

- **Parsing accuracy risk:** Resumes vary widely in format; text extraction may misinterpret complex layouts. Mitigation: test against diverse sample templates early and provide manual correction fields.
- **AI fabrication risk:** LLM rewriter could hallucinate experience. Mitigation: constrain prompts strictly and enforce a validation check flagging any unauthorized terms.

- **API cost/availability risk:** Third-party LLM rate limits or costs. Mitigation: implement caching for repeated calls and fall back to lighter/open models during development.
- **Scope risk:** Combining 8 features in one semester. Mitigation: maintain a clean single-database architecture and reserve buffer weeks for refinement.

11 Summary

This project proposes a deployable web platform for an AI-Powered ATS Resume Scanner and Job-Readiness Assistant. It adheres to all course criteria by prioritizing time-to-value through rapid iterations, employing automated testing for reliable delivery, and defining measurable, explainable evaluation metrics. The platform connects resume matching, gap analysis, bullet enhancement, and role-specific mock interviewing into a single unified workflow for job seekers.